Assignment 5

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Information Gathering

Objective: Perform active and passive reconnaissance of the assigned target, identify exposed ports and services, and answer the five questions provided in the TuteDude TryHackMe room.

1. Methodology

Room connection: The assigned TuteDude-CyberSecurity TryHackMe environment was accessed and the target machine information was obtained before reconnaissance.

Active reconnaissance: Nmap service/version detection was used to identify open TCP ports and running services using `nmap -sV` and `nmap -sC -sV`.

Passive reconnaissance: Public domain-registration information was used to determine the registration date of `tutedude.com`, followed by verification in the TryHackMe room.

2. Target Connection Evidence

```
-------------------------------------------------
WELCOME TO THE CyberSecurity CHALLENGE
-------------------------------------------------

TARGET IP ADDRESS: 2409:4091:8001:7e04:20c:29ff:fe06:2d66
WEB PORT: 8080

Login as 'student:student123' via SSH once you have credentials.
-------------------------------------------------
hacking login: student
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-101-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Oct  6 04:06:52 PM UTC 2026

  System load:            0.15
  Usage of /:             38.3% of 9.75GB
  Memory usage:           16%
  Swap usage:             0%
  Processes:              259
  Users logged in:        0
  IPv4 address for ens33: 192.168.31.236
  IPv6 address for ens33: 2409:4091:8001:7e04:20c:29ff:fe06:2d66

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

74 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

student@hacking:~$
```

Figure 1. Target machine information and SSH connection.

3. Active Reconnaissance – Nmap

The Nmap results identified the target's exposed services, including FTP, SSH, HTTP, NetBIOS/SMB, and an additional HTTP service on port 8080.

```
┌──(sai㉿kali)-[~]
└─$ nmap -sV 192.168.31.236
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-07 00:07 +0800
Nmap scan report for hacking.lan (192.168.31.236)
Host is up (0.00018s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 3.0.5
22/tcp   open  ssh         OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
80/tcp   open  http        Apache httpd 2.4.58 ((Ubuntu))
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
8080/tcp open  http        Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 00:0C:29:06:2D:66 (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.58 seconds
```

Figure 2. Nmap service/version scan.

```
┌──(sai㉿kali)-[~]
└─$ nmap -sC -sV 192.168.31.236
Starting Nmap 7.98 ( https://nmap.org ) at 2026-10-07 00:07 +0800
Nmap scan report for hacking.lan (192.168.31.236)
Host is up (0.00018s latency).
Not shown: 994 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 3.0.5
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_-rw-r--r--    1 0        0              26 Feb 20  2026 test.txt
| ftp-syst:
|   STAT:
| FTP server status:
|      Connected to ::ffff:192.168.31.137
|      Logged in as ftp
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 1
|      vsFTPd 3.0.5 - secure, fast, stable
|_End of status
22/tcp   open  ssh         OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 26:ed:49:c2:2b:1e:8e:a2:4a:c7:75:65:28:c7:35:4f (ECDSA)
|_  256 2f:27:b2:c7:7b:29:d3:84:e1:3a:94:61:cc:35:ba:bf (ED25519)
80/tcp   open  http        Apache httpd 2.4.58 ((Ubuntu))
| http-robots.txt: 3 disallowed entries
|_/backups/ /config/ /tmp/
|_http-server-header: Apache/2.4.58 (Ubuntu)
|_http-title: SecureCorp | Employee Portal
| http-cookie-flags:
|   /:
|     PHPSESSID:
|_      httponly flag not set
139/tcp  open  netbios-ssn Samba smbd 4
445/tcp  open  netbios-ssn Samba smbd 4
8080/tcp open  http        Apache httpd 2.4.58 ((Ubuntu))
|_http-open-proxy: Proxy might be redirecting requests
| http-title: Sentinel Portal - Legacy Management System
|_Requested resource was login.php
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 00:0C:29:06:2D:66 (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Host script results:
|_nbstat: NetBIOS name: HACKING, NetBIOS user: <unknown>, NetBIOS MAC: <unknown> (unknown)
| smb2-time:
|   date: 2026-10-06T16:08:08
|_  start_date: N/A
| smb2-security-mode:
|   3.1.1:
|_    Message signing enabled but not required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.21 seconds
```

Figure 3. Detailed Nmap scan with service information.

4. Findings and Answers

1. Domain registration date: `22/11/2019`
2. Number of open ports: 7
3. Common SSH port: 22
4. Service running on port 80: `http`
5. Service/version running on port 21: `vsftpd 3.0.5`

The service and port information was obtained through Nmap, and the final values were submitted to the assigned TryHackMe room for verification.

5. TryHackMe Verification

All five answers were submitted in the assigned TryHackMe room and were confirmed with **Correct Answer** status.



Figure 4. TryHackMe questions showing all five answers as correct.

6. Conclusion

The exercise demonstrated both passive and active reconnaissance. Passive reconnaissance was used to obtain domain-registration information, while Nmap was used for active service and port enumeration. The resulting information was used to answer the five assigned questions, and the answers were successfully verified in the TryHackMe room.

Final verified values: 22/11/2019 | 7 open ports | SSH 22 | HTTP on port 80 | vsftpd 3.0.5 on port 21.